```
!pip install pgmpy
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.inference import VariableElimination
import pandas as pd

# Step 1: Dataset (Loan Data)
data = pd.DataFrame({
    'Income': ['Stable', 'Stable', 'Unstable', 'Stable', 'Unstable', 'Unstable', 'Stable', 'Unstable'],
    'Credit': ['Good', 'Bad', 'Good', 'Good', 'Bad', 'Bad', 'Bad', 'Good'],
    'Default': ['No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No', 'Yes']
})

# Step 2: Define Bayesian Network
model = DiscreteBayesianNetwork([
    ('Income', 'Default'),
    ('Credit', 'Default')
])

# Step 3: Train Model (MLE)
model.fit(data)

# Step 4: Print CPDs
print("\n--- CPDs ---")
for cpd in model.get_cpds():
    print(cpd)

# Step 5: Inference
inference = VariableElimination(model)

# Step 6: Query (incomplete data)
result = inference.query(
    variables=['Default'],
    evidence={'Income': 'Stable'}
)

# Step 7: Output
print("\n--- Result ---")
print(result)
```

```
Collecting pgmpy
  Downloading pgmpy-1.1.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: huggingface_hub>=0.23 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.7.1)
Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (3.6.1)
Requirement already satisfied: numpy>=2.0 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (2.0.2)
Requirement already satisfied: scipy>=1.10 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.16.3)
Requirement already satisfied: scikit-learn>=1.2 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.6.1)
Requirement already satisfied: pandas>=1.5 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (2.2.2)
Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (0.14.6)
Requirement already satisfied: tqdm>=4.64 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (4.67.3)
Requirement already satisfied: pyparsing>=3.0 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (3.3.2)
Requirement already satisfied: joblib>=1.2 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.5.3)
Requirement already satisfied: opt_einsum>=3.3 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (3.4.0)
Collecting scikit-base>=0.12.4 (from pgmpy)
  Downloading scikit_base-0.13.1-py3-none-any.whl.metadata (8.8 kB)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgr
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy)
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy) (6.0
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5->pgmpy) (2.9
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5->pgmpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5->pgmpy) (2025.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.2->pgmpy)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels>=0.14.5->pgmpy) (1.0.
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.23-
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.2
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hul
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.23->p
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hug
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.5->p
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub>=0.23->pgmp
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub>=0.2
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub>=0.23->pgmp
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface_hub>=0
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
```

```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hu
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0
Downloading pgmpy-1.1.0-py3-none-any.whl (2.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.4/2.4 MB 36.2 MB/s eta 0:00:00
Downloading scikit_base-0.13.1-py3-none-any.whl (159 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 159.8/159.8 kB 8.8 MB/s eta 0:00:00
Installing collected packages: scikit-base, pgmpy
Successfully installed pgmpy-1.1.0 scikit-base-0.13.1
/usr/local/lib/python3.12/dist-packages/pgmpy/estimators/__init__.py:4: FutureWarning: `pgmpy.estimators.StructureScore` is dep
  from .StructureScore import (

--- CPDs ---
+------------------+-----+
| Income(Stable)   | 0.5 |
+------------------+-----+
| Income(Unstable) | 0.5 |
+------------------+-----+
+--------------+----------------+-----+------------------+
```